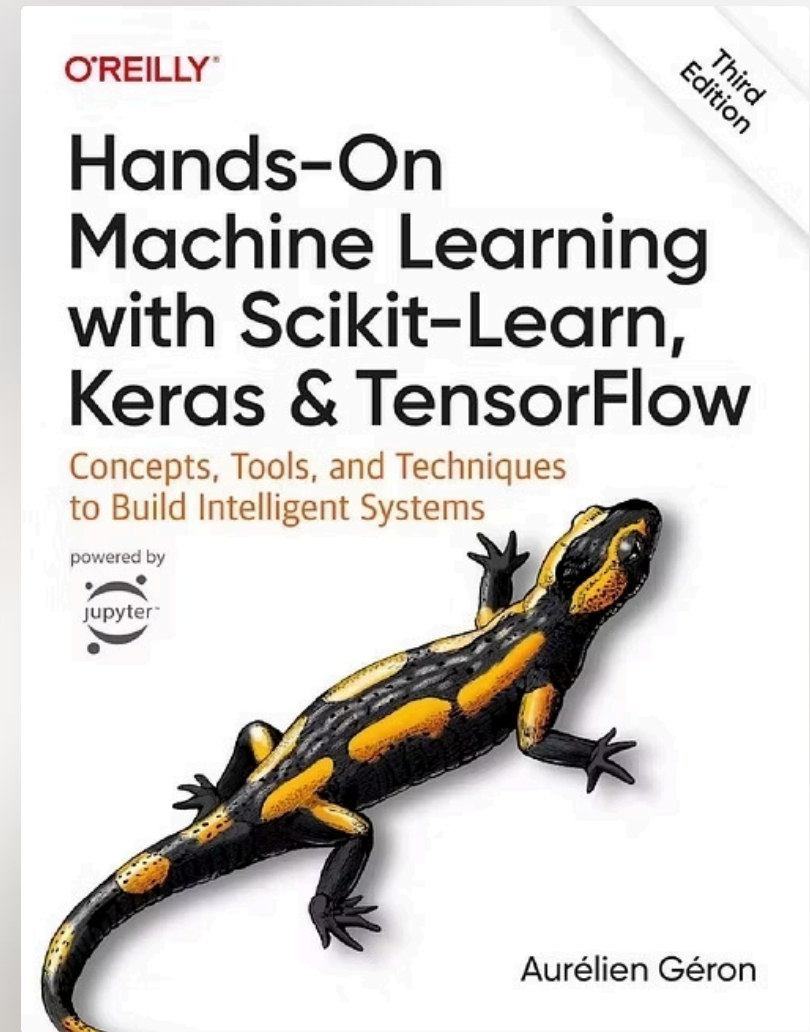


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 07/01/2025



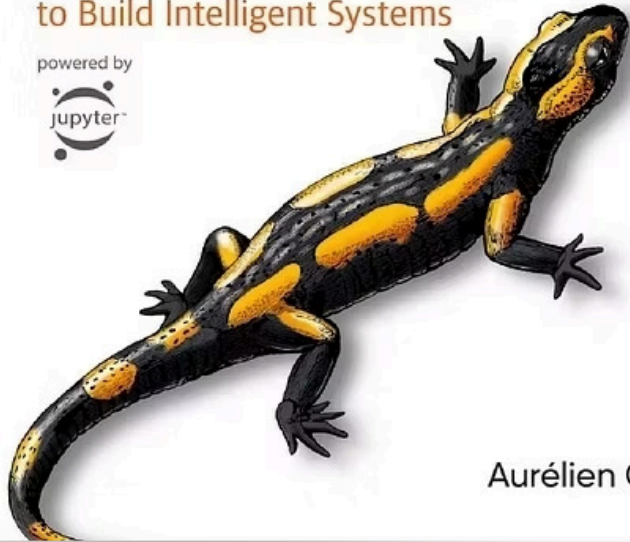
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Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
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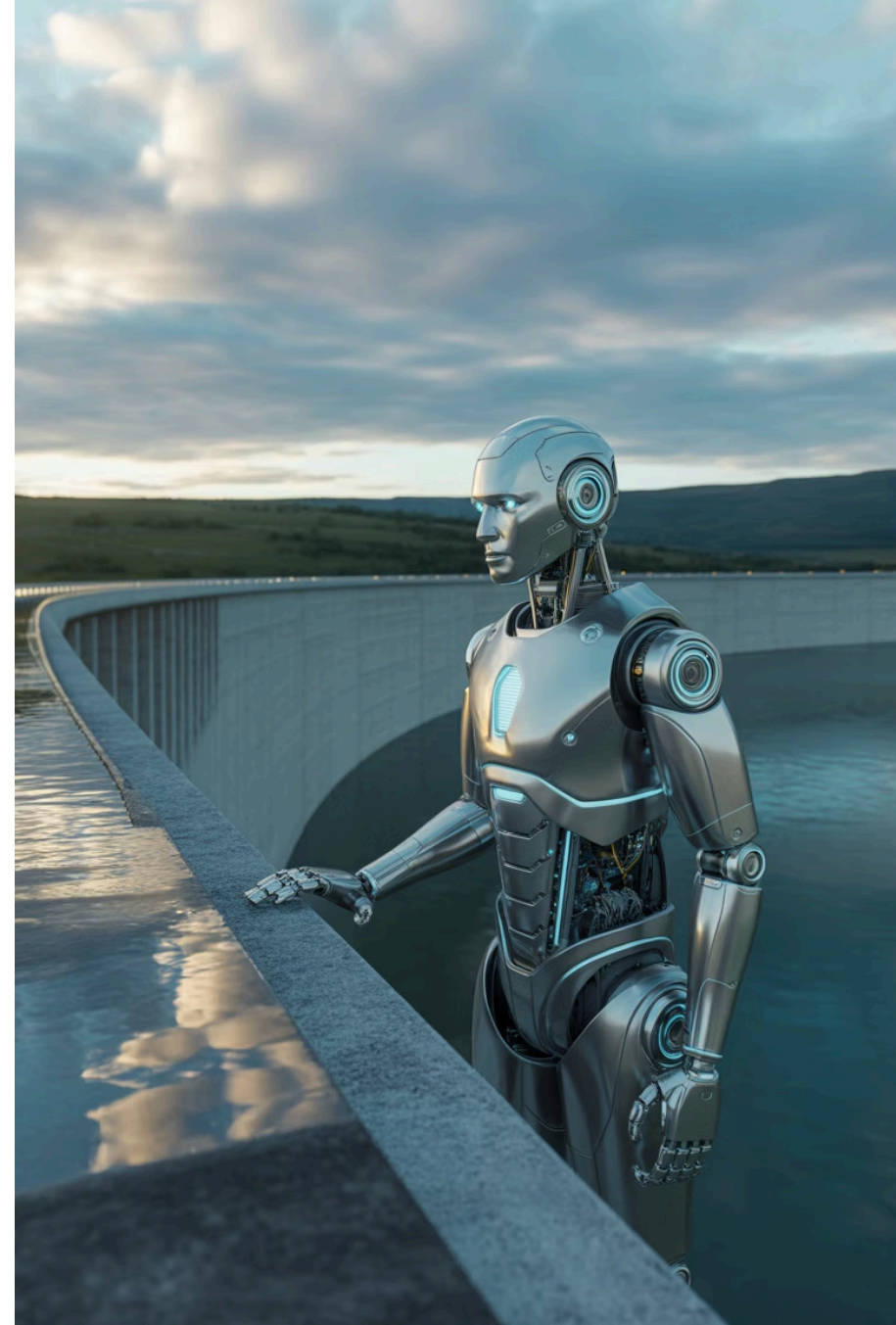
Aurélien Géron

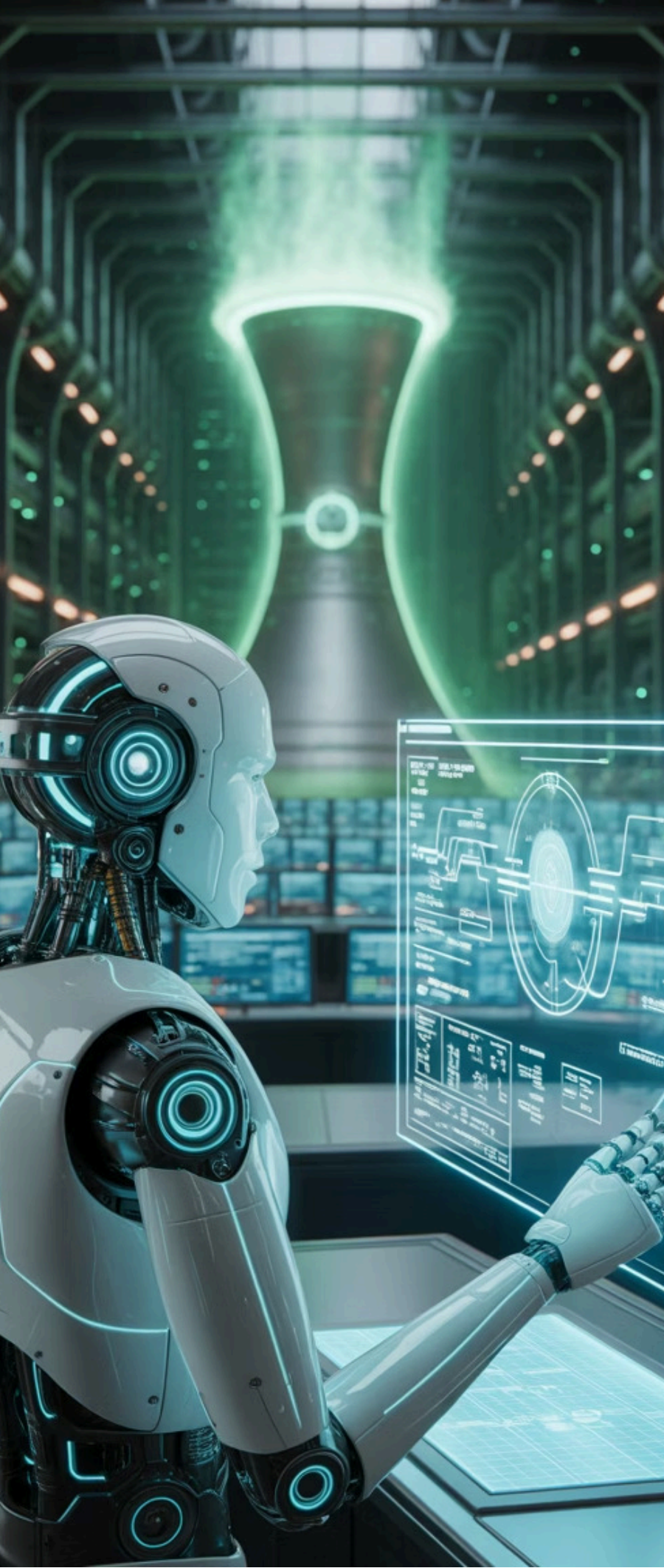
Machine Learning

Chapter 1 – The Machine Learning Landscape

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
- 4. Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. Testing and Validating

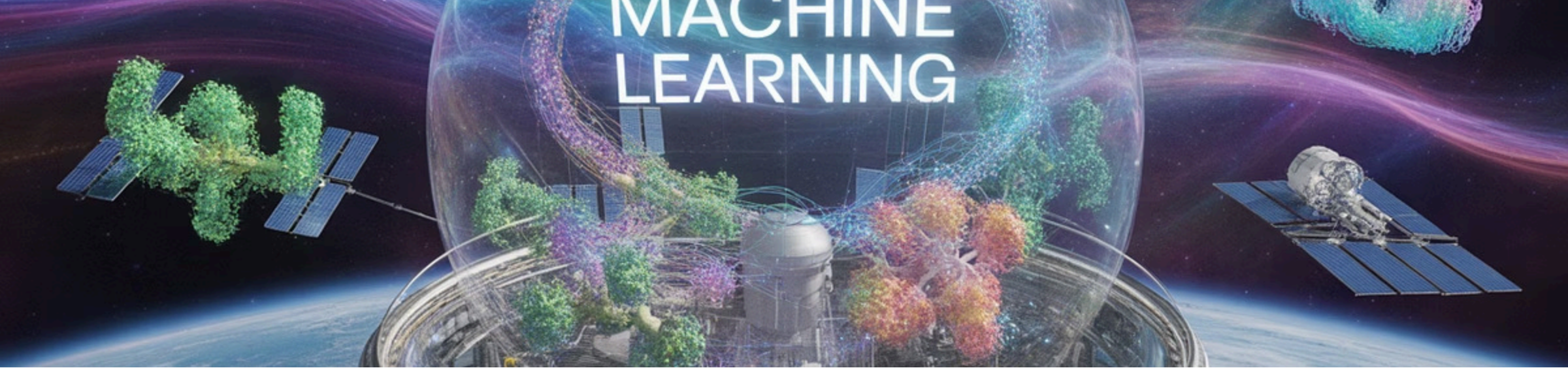




Agenda for Chapter 1

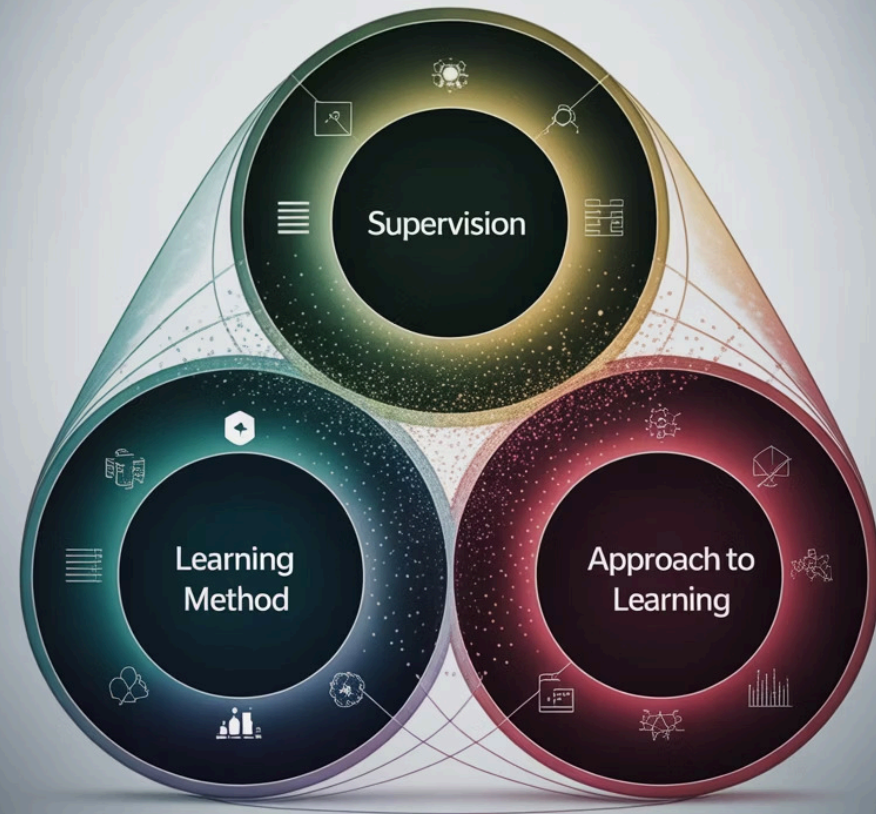
1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
 - **Classification criteria #1:** Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - **Classification criteria #2:** Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - **Classification criteria #3:** Learning Method
 - Instance-Based and Model-Based Learning
5. **Main Challenges of Machine Learning**
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back
6. **Testing and Validating**
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch

Types of Machine Learning Systems



There are so many different types of ML systems that it is useful to classify them in broad categories, based some criteria.

Machine Learning Classification Criteria



Three Classification Criteria for ML Systems

1. Supervision during training

- Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning

2. Learning method

- Batch and Online (Incremental) Learning

3. Approach to learning

- Instance-based and model-based Learning



Classification criteria #1: Supervision during training

Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning

Classification criteria #1: Supervision during training



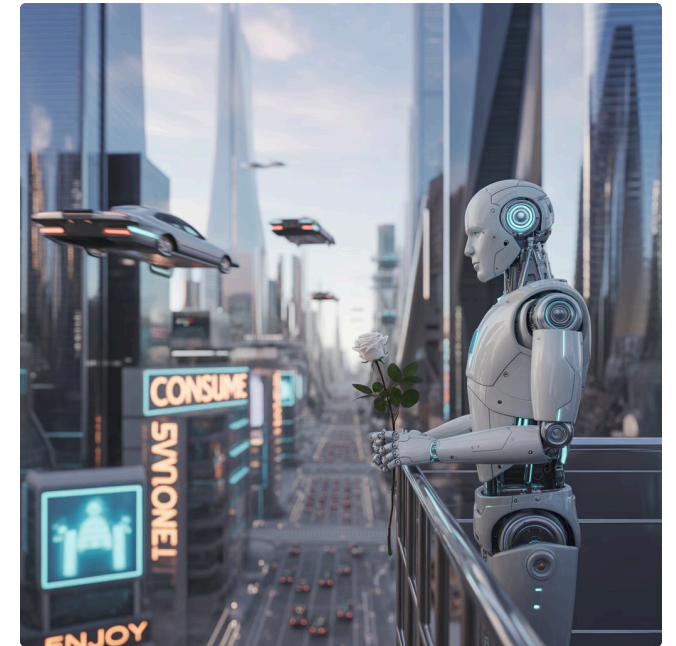
Supervised Learning

- Requires labeled data
- **Examples:** Classification, regression



Unsupervised Learning

- No labeled data required
- **Examples:** Clustering, association



Semi-Supervised Learning

- Uses both labeled and unlabeled data
- **Examples:** Image classification with limited labels

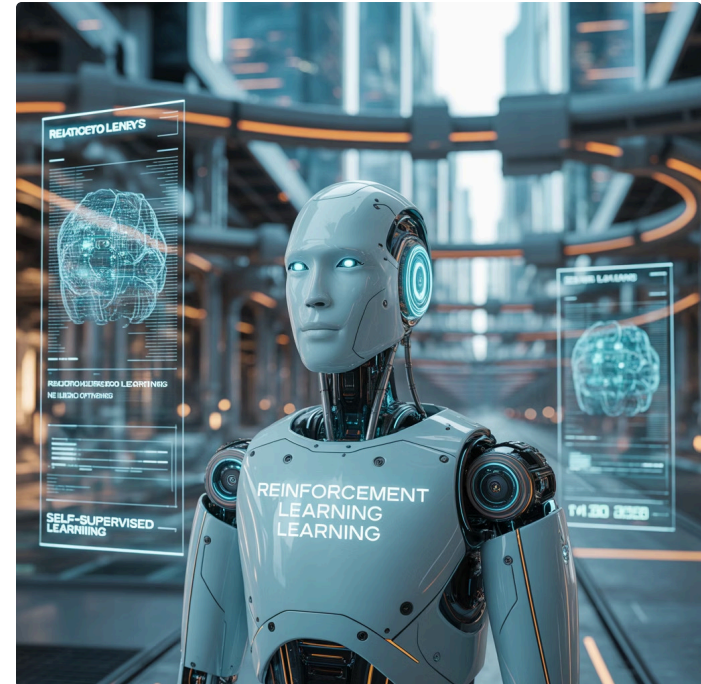
Classification criteria #1: Supervision during training (cont.)



Self-Supervised Learning

- **Generates labels automatically from unlabeled data**
- **Often used for pre-training in deep learning models**
- **Examples:** Predicting missing words in sentences (e.g., BERT*), image completion

*BERT - Bidirectional Encoder Representations from Transformers



Reinforcement Learning

- **Learns through trial and error by interacting with an environment**
- **Receives feedback in the form of rewards or penalties**
- **Examples:** Game playing, robotics



Classification criteria #2: Approach to Learning Method

Batch and Online (Incremental) Learning

Batch vs. Online Learning



Batch Learning:

Learns in large batches, periodically updated

Cannot adapt in real-time



Online Learning:

Learns incrementally, one data point at a time

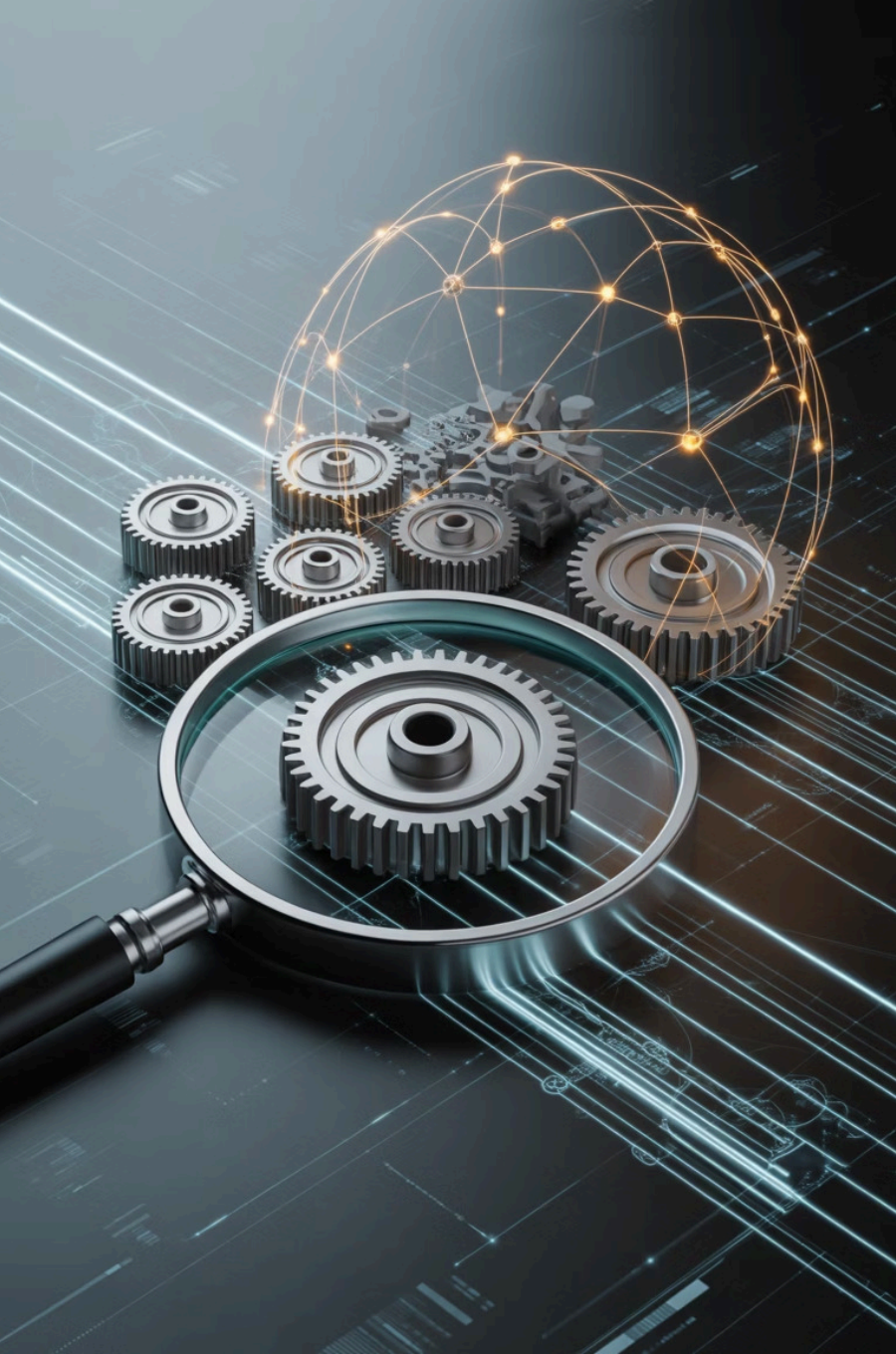
Adapts to new data in real-time





Classification criteria #3: Learning Method

Instance-based and model-based learning



Instance-Based vs. Model-Based Learning

Instance-Based Learning:

Compares new data to known examples directly

Example: k-Nearest Neighbors (KNN)

Model-Based Learning:

Detects patterns and builds a predictive model

Example: Decision Trees, Neural Networks

Example: A Spam Filter

These criteria are not exclusive; you can combine them in any way you like.



Spam Filter:

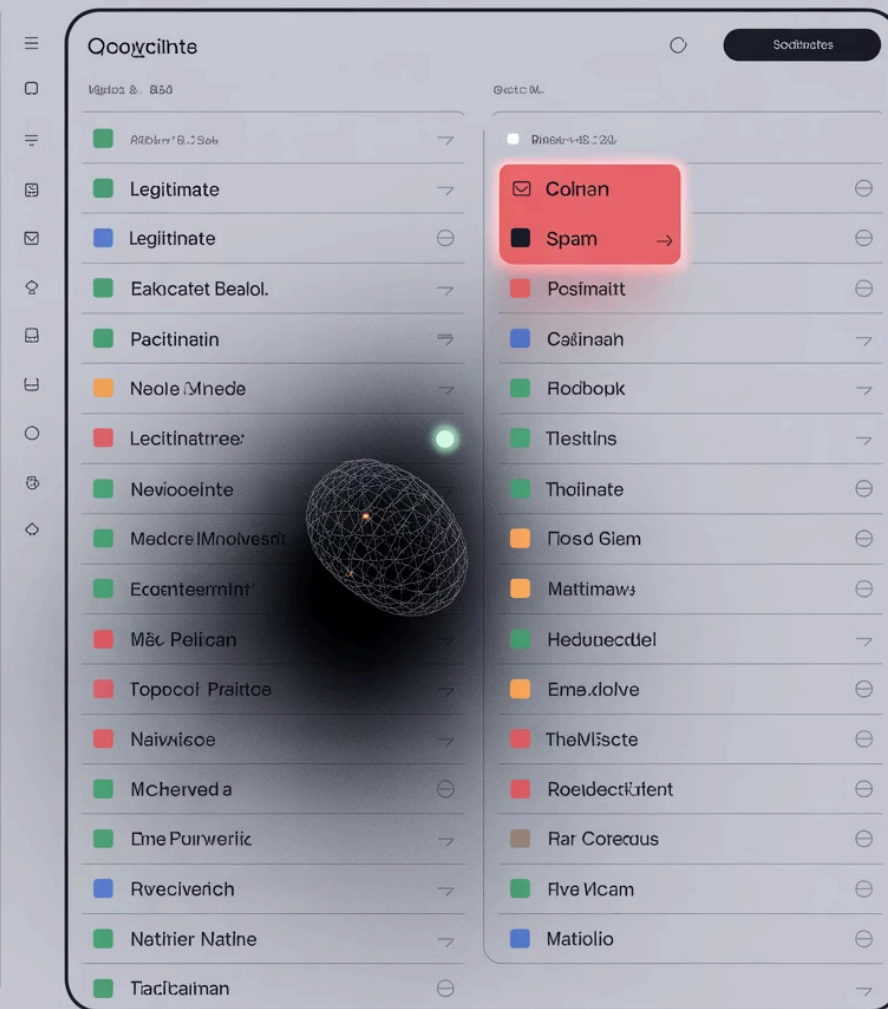
- **Learning type:** Online, model-based, supervised learning system
- **How it works:** Learns incrementally using human-provided spam/ham examples
- **Method:** Uses a deep neural network model

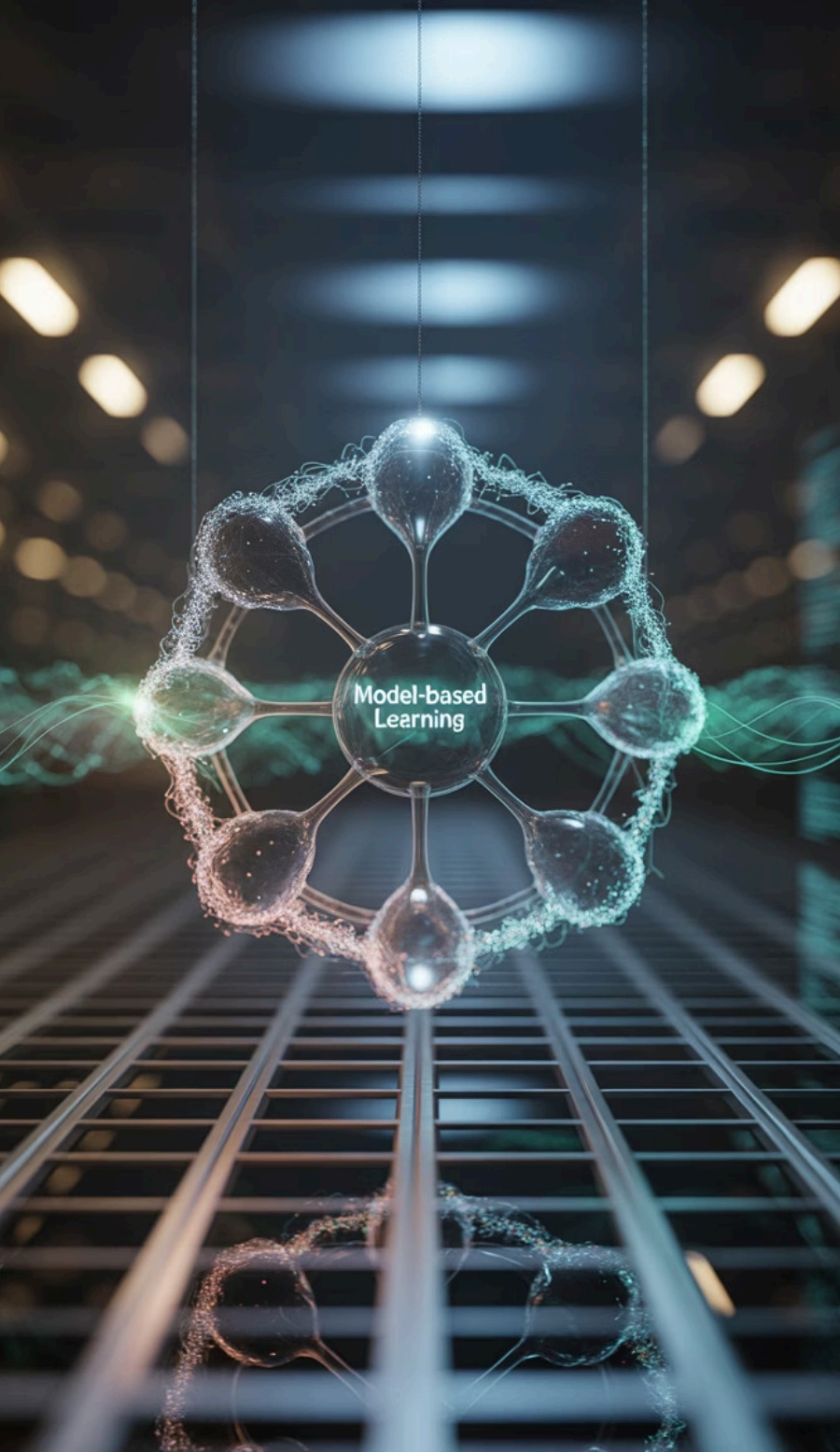
In the next lessons of this chapter we are going to take a look at each of these criteria a bit more closely.

Email Insights

Settings

Inbox Sent Sent Drafts Spam



A futuristic, glowing neural network structure. A central node is labeled "Model-based Learning". The structure is composed of interconnected nodes and edges, with a central node labeled "Model-based Learning". The background is dark with blue and green light effects, suggesting a high-tech or artificial intelligence theme.

Why the Spam Filter example is model-based (and not instance-based):

In **model-based learning**, the system builds a general **model** (like a neural network) from the **training data**.

- Once trained, this **model summarizes patterns** and is used to make **predictions on new data** without needing to **store or compare to all training instances**.

Instance-based learning (like k-Nearest Neighbors) **stores the training data itself** and makes **predictions by directly comparing new examples to stored examples**.

Model vs. Instance-Based Learning

Aspect	Model-Based Learning	Instance-Based Learning
How it works	Learns a general model (e.g., neural network, decision tree) from training data	Stores all or many training instances and uses them to predict new cases
Prediction	Uses the model's parameters to make predictions	Compares new examples directly to stored instances (e.g., nearest neighbors)
Spam Filter example	Builds a deep neural network to classify emails	Would keep many labeled emails and compare new emails to them directly
Pros	Fast predictions; generalizes well; compact representation	Can adapt quickly; works well with complex patterns directly from data
Cons	May require careful training and tuning; risks over/underfitting	Slow prediction time (must compare to many instances); needs lots of memory
Book's spam filter	✅ Uses a deep learning model to generalize spam vs. ham	❌ Does <i>not</i> store old emails for direct comparison



Next: Supervised Learning

Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts